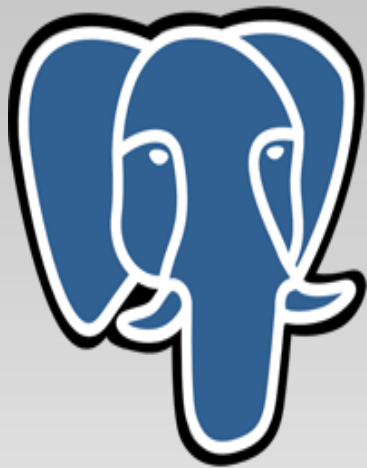




PostgreSQL

POSTGRES DATABASE ADMINISTRATION



STEPHEN NJOROGUE

LinkedIn: www.linkedin.com/in/stephen-njoroge

|SENIOR DATABASE ADMINISTRATOR|ORACLE| POSTGRESQL | MSSQL|OCI|AWS|

CHAPTER 1: POSTGRESQL SYSTEM ARCHITECTURE

About the Author:

I am Stephen Njoroge, a Senior Database Administrator with expertise in **MSSQL, Oracle and PostgreSQL**, and strong proficiency in **OCI, AWS and Proxmox** for cloud and virtualization solutions. I'm skilled in **Prometheus, Grafana and iTop** for monitoring and IT operations as well as **Tableau, Power BI and Oracle BI** for advanced data analytics and visualization.

CHAPTER 1

PostgreSQL System Architecture

In this chapter, we learn about the origin of PostgreSQL, its architecture, and its Components and understand the various components of PostgreSQL during a client Connection request.

We cover the following topics:

- PostgreSQL key architectural components
- PostgreSQL server process tree
- PostgreSQL architecture flow

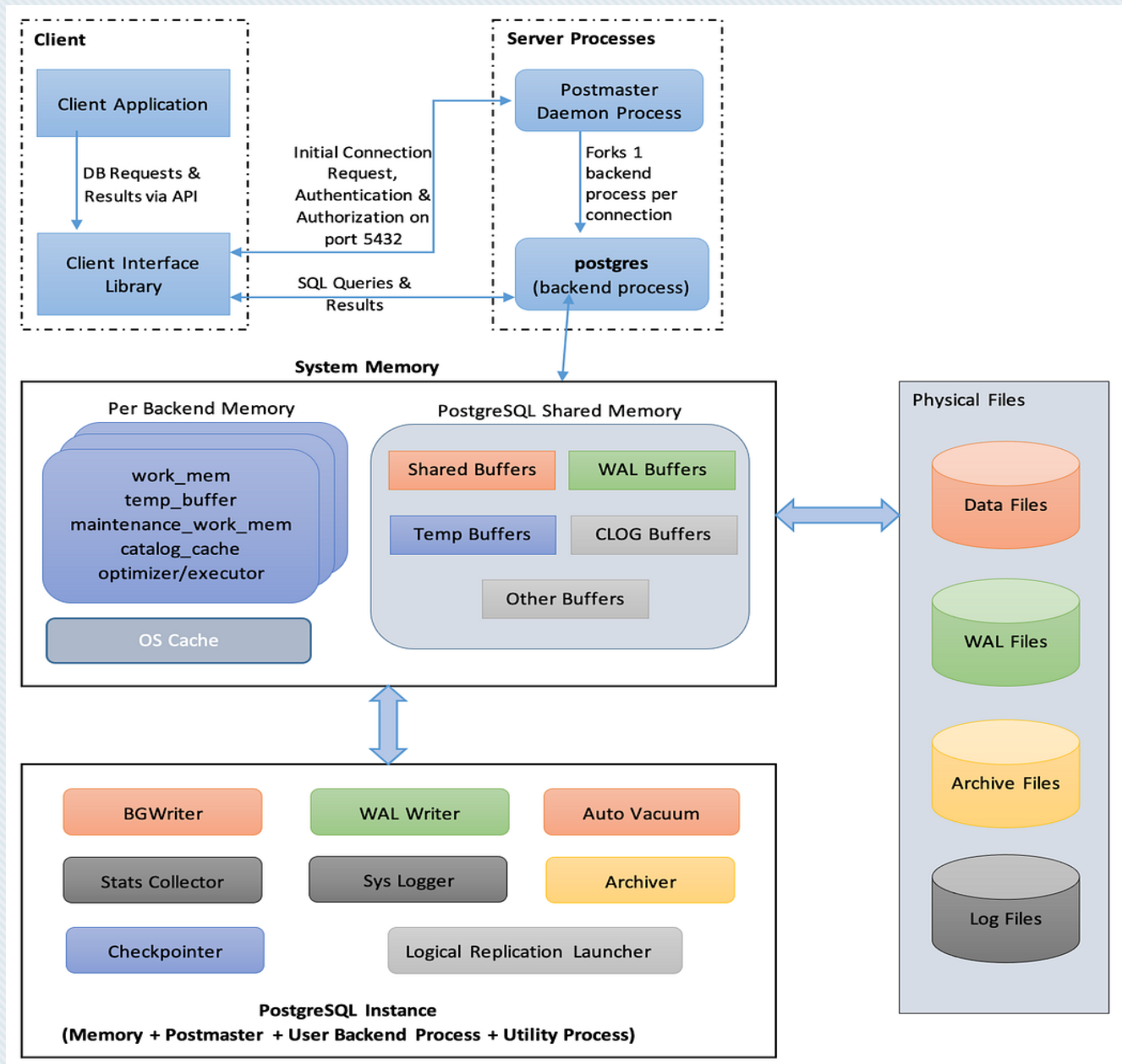
PostgreSQL Components

The key components in a PostgreSQL architecture are

- Server process
- Utility process
- System memory
- Physical files

Each of these core components further relies on several other components as shown below.

PostgreSQL System Architecture



CLIENT LAYER COMPONENTS

- Client Application

Any application that communicates with PostgreSQL (web apps, desktop apps, scripts, BI tools).

-- View client application names from active connections

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```
SELECT pid, username, application_name, client_addr
FROM pg_stat_activity
WHERE application_name != '';
```

pid	username	application_name	client_addr
3708	postgres	psql	
3768	training	psql	127.0.0.1

(2 rows)

training=#

- Client Interface Library

Database driver/library (libpq, psycopg2, JDBC, etc.) that implements PostgreSQL protocol.

```
# Check library versions (OS level)
```

```
ldconfig -p | grep pq
```

```
root@998-986-02:/home/steve# ldconfig -p | grep pq
libpq.so.5 (libc6,x86-64) => /lib/x86_64-linux-gnu/libpq.so.5
root@998-986-02:/home/steve#
```

```
# Python:
```

```
Python3 -c "import psycopg2; print(psycopg2.__version__)"
```

```
root@998-986-02:/home/steve# python3 -c "import psycopg2; print(psycopg2.__version__)"
2.9.9 (dt dec pq3 ext lo64)
root@998-986-02:/home/steve#
```

- PostgreSQL Protocol

Binary protocol used for client-server communication (default port 5432).

```
# Check listening port
```

```
netstat -tlnp | grep 5432
```

```
root@998-986-02:/home/steve# netstat -tlnp | grep 5432
tcp        0      0 127.0.0.1:5432      0.0.0.0:*           LISTEN      3650/postgres
root@998-986-02:/home/steve#
```

```
ss -tlnp | grep 5432
```

```
root@998-986-02:/home/steve# ss -tlnp | grep 5432
LISTEN 0      200      127.0.0.1:5432      0.0.0.0:*           users:(("postgres",pid=3650,fd=6))
root@998-986-02:/home/steve#
```

```
# In PostgreSQL:
SHOW port;
```

```
training=# SHOW port;
 port
-----
 5432
(1 row)

training=#
```

SERVER PROCESSES

- Postmaster Daemon Process

Master process that starts PostgreSQL, allocates shared memory, and manages connections.

```
# Find postmaster PID and details
sudo head -1 /var/run/postgresql/16-main.pid
```

```
root@998-986-02:/home/steve# sudo head -1 /var/run/postgresql/16-main.pid
3650
root@998-986-02:/home/steve#
```

```
#
ps -ef | grep postgres | grep -v grep
```

```

root@998-986-02:/home/steve# ps -ef | grep postgres | grep -v grep
postgres 3650      1  0 02:54 ?        00:00:06 /usr/lib/postgresql/16/bin/postgres -D /var/lib/postgresql/16/main -c config_file
=/etc/postgresql/16/main/postgresql.conf
postgres 3651    3650  0 02:54 ?        00:00:00 postgres: 16/main: checkpointer
postgres 3652    3650  0 02:54 ?        00:00:00 postgres: 16/main: background writer
postgres 3654    3650  0 02:54 ?        00:00:00 postgres: 16/main: walwriter
postgres 3655    3650  0 02:54 ?        00:00:00 postgres: 16/main: autovacuum launcher
postgres 3656    3650  0 02:54 ?        00:00:00 postgres: 16/main: logical replication launcher
root      3703      848  0 02:55 pts/4    00:00:00 sudo -u postgres psql
root      3704    3703  0 02:55 pts/5    00:00:00 sudo -u postgres psql
postgres 3705    3704  0 02:55 pts/5    00:00:00 /usr/lib/postgresql/16/bin/psql
postgres 3708    3650  0 02:56 ?        00:00:00 postgres: 16/main: postgres training [local] idle
root@998-986-02:/home/steve#

```

```

-- Check if postmaster is running
SELECT pg_is_in_recovery();

```

```

training=# SELECT pg_is_in_recovery();
 pg_is_in_recovery
-----
 f
(1 row)

training=#

```

- Backend Process (postgres)

Child process created for each client connection, handles query execution.

```

-- View all backend processes
SELECT pid, username, datname, state, query_start, query
FROM pg_stat_activity
WHERE backend_type = 'client backend';

```

```

WHERE backend_type = 'client backend'
 pid | username | datname | state | query_start | query
-----+-----+-----+-----+-----+-----
 3708 | postgres | training | active | 2025-12-14 10:30:07.126442+03 | SELECT pid, username, datname, state, query_start, query +
 | FROM pg_stat_activity | FROM pg_stat_activity +
 | WHERE backend_type = 'client backend'; | WHERE backend_type = 'client backend';
(1 row)

training=#

```

OS view

```
ps aux | grep "postgres.*postgres" | grep -v "grep\|master"
```

```

root@998-986-02:/home/steve# ps aux | grep "postgres.*postgres" | grep -v "grep\|master"
postgres 3650  0.0  0.7 220244 30796 ?        Ss   02:55   0:06 /usr/lib/postgresql/16/bin/postgres -D /var/lib/postgresql/16/main
-c config_file=/etc/postgresql/16/main/postgresql.conf
postgres 3651  0.0  0.5 220532 22068 ?        Ss   02:55   0:00 postgres: 16/main: checkpointer
postgres 3652  0.0  0.1 220396 5828 ?        Ss   02:55   0:00 postgres: 16/main: background writer
postgres 3654  0.0  0.2 220244 10028 ?        Ss   02:55   0:00 postgres: 16/main: walwriter
postgres 3655  0.0  0.2 221836 8428 ?        Ss   02:55   0:00 postgres: 16/main: autovacuum launcher
postgres 3656  0.0  0.1 221820 6692 ?        Ss   02:55   0:00 postgres: 16/main: logical replication launcher
postgres 3705  0.0  0.2 20904 8908 pts/5    S+   02:55   0:00 /usr/lib/postgresql/16/bin/psql
postgres 3708  0.0  0.5 223824 20824 ?        Ss   02:56   0:00 postgres: 16/main: postgres training [local] idle
root@998-986-02:/home/steve#

```

- Background Writer (BGWriter)

Writes dirty buffers from shared buffers to disk periodically.

```
-- Check BGWriter statistics
```

```
SELECT
    checkpoints_timed,
    checkpoints_req,
    checkpoint_write_time,
    checkpoint_sync_time
FROM pg_stat_bgwriter;
```

```
checkpoints_timed | checkpoints_req | checkpoint_write_time | checkpoint_sync_time
-----+-----+-----+-----
          92 |           0 |         105460 |           44
(1 row)
```

```
-- Configuration
```

```
SHOW bgwriter_delay;
```

```
training=# SHOW bgwriter_delay;
bgwriter_delay
-----
    200ms
(1 row)
```

```
training=#
```

```
SHOW bgwriter_lru_maxpages;
```

```
training=# SHOW bgwriter_lru_maxpages;
bgwriter_lru_maxpages
-----
         100
(1 row)

training=#
```

- Checkpointer

Performs checkpoints - ensures all dirty pages are written to disk at consistent points.

```
# Process view
```

```
ps aux | grep "postgres.*checkpointer" | grep -v grep
```

```
root@998-986-02:/home/steve# ps aux | grep "postgres.*checkpointer" | grep -v grep
postgres   3651  0.0  0.5 220532 22068 ?        Ss   02:56   0:00 postgres: 16/main: checkpointer
root@998-986-02:/home/steve#
```

```
-- Checkpoint configuration
```

```
SHOW checkpoint_timeout;
```

```
training=# SHOW checkpoint_timeout;
checkpoint_timeout
-----
         5min
(1 row)
```

```
training=#
```

```
SHOW checkpoint_completion_target;
```

```
training=# SHOW checkpoint_completion_target;
checkpoint_completion_target
-----
0.9
(1 row)
```

```
training=#
```

```
SHOW max_wal_size;
```

```
training=# SHOW max_wal_size;
max_wal_size
-----
1GB
(1 row)
```

```
training=# |
```

```
-- Last checkpoint time
```

```
SELECT stats_reset AS last_checkpoint_stats_reset
FROM pg_stat_bgwriter;
```

```
last_checkpoint_stats_reset
-----
2025-12-13 19:35:05.209222+03
(1 row)
```

```
training=#
```

- WAL Writer

Writes Write-Ahead Log records from WAL buffers to WAL files.

```
ps aux | grep "postgres.*walwriter" | grep -v grep
```

```
root@998-986-02:/home/steve# ps aux | grep "postgres.*walwriter" | grep -v grep
postgres    3654  0.0  0.2 220244 10028 ?        Ss   02:57   0:00 postgres: 16/main: walwriter
root@998-986-02:/home/steve#
```

```
-- WAL writer configuration
```

```
SHOW wal_writer_delay;
```

```
training=# SHOW wal_writer_delay;
wal_writer_delay
-----
200ms
(1 row)
```

```
training=#
```

```
SHOW wal_writer_flush_after;
```

```
training=# SHOW wal_writer_flush_after;
 wal_writer_flush_after
-----
 1MB
(1 row)
```

```
training=#
```

```
-- Current WAL position
SELECT pg_current_wal_lsn();
```

```
training=# SELECT pg_current_wal_lsn();
 pg_current_wal_lsn
-----
 0/195C480
(1 row)
```

```
training=#
```

- Autovacuum Launcher

Manages autovacuum worker processes for cleaning up dead tuples.

```
ps aux | grep "postgres.*autovacuum launcher" | grep -v grep
```

```
root@998-986-02:/home/steve# ps aux | grep "postgres.*autovacuum launcher" | grep -v grep
postgres    3655  0.0  0.2 221836 8428 ?        Ss   02:58   0:00 postgres: 16/main: autovacuum launcher
root@998-986-02:/home/steve#
```

```
-- Autovacuum settings
```

```
SHOW autovacuum;
```

```
SHOW autovacuum_max_workers;
```

```
SHOW autovacuum_naptime;
```

```

training=# SHOW autovacuum;
autovacuum
-----
on
(1 row)

training=# SHOW autovacuum_max_workers;
autovacuum_max_workers
-----
3
(1 row)

training=# SHOW autovacuum_naptime;
autovacuum_naptime
-----
1min
(1 row)

```

- Archiver

Archives WAL files to a backup location for PITR (Point-in-Time Recovery).

```
ps aux | grep "postgres.*archiver" | grep -v grep
```

-- *Archive settings*

```

SHOW archive_mode;
SHOW archive_command;

```

```

postgres=# SHOW archive_mode;
SHOW archive_command;
archive_mode
-----
on
(1 row)

                                archive_command
-----
test ! -f /var/lib/postgresql/16/archive/%f && cp %p /var/lib/postgresql/16/archive/%f
(1 row)

postgres=# |

```

- Statistics Collector

Collects database usage statistics for query optimization.

```

-- Statistics settings
SHOW track_activities;
SHOW track_counts;
SHOW track_io_timing;

```

```

postgres=# SHOW track_activities;
track_activities
-----
on
(1 row)

postgres=# SHOW track_counts;
track_counts
-----
on
(1 row)

postgres=# SHOW track_io_timing;
track_io_timing
-----
off
(1 row)

postgres=#

```

- Log Writer (Sys Logger)

Writes log messages to log files.

```

-- Logging configuration
SHOW logging_collector;
SHOW log_destination;
SHOW log_directory;
SHOW log_filename;
-- Current log file
SELECT pg_current_logfile();

```

```

SELECT pg_current_logfile();
 logging_collector
-----
off
(1 row)

log_destination
-----
stderr
(1 row)

log_directory
-----
log
(1 row)

log_filename
-----
postgresql-%Y-%m-%d_%H%M%S.log
(1 row)

pg_current_logfile
-----
(1 row)
postgres=#

```

- Logical Replication Launcher

Manages logical replication workers for logical decoding.

```

-- Logical replication settings
SHOW max_logical_replication_workers;
SHOW max_sync_workers_per_subscription;

```

```

postgres=# SHOW max_logical_replication_workers;
SHOW max_sync_workers_per_subscription;
 max_logical_replication_workers
-----
4
(1 row)

 max_sync_workers_per_subscription
-----
2
(1 row)
postgres=#

```

- Startup Process

Handles database recovery during startup.

```
-- Check if in recovery
SELECT pg_is_in_recovery();
```

```
postgres=# SELECT pg_is_in_recovery();
pg_is_in_recovery
-----
f
(1 row)

postgres=#
```

- WAL Sender/Receiver Processes

Handle streaming replication.

```
-- WAL senders (primary)
SELECT * FROM pg_stat_replication;
```

```
application_name | client_addr | client_hostname | client_port
-----+-----+-----+-----
(0 rows)

postgres=# |
```

```
-- WAL receivers (standby)
SELECT * FROM pg_stat_wal_receiver;
```

```
pid | status | receive_start_lsn | receive_start_tli | written_lsn | flushed_lsn | received_tli
-----+-----+-----+-----+-----+-----+-----
(0 rows)

postgres=# |
```

SYSTEM MEMORY COMPONENTS

- Shared Memory (Allocated at startup)
- Shared Buffers

Cache for database pages (tables, indexes) in memory.

```
-- Configuration
SHOW shared_buffers;
-- Usage statistics
SELECT
    pg_size_pretty(count(*) * 8192) as buffer_cache_size,
    round(100.0 * sum(CASE WHEN isdirty THEN 1 ELSE 0 END) / count(*), 2) as percent_dirty
FROM pg_buffercache;
```

```

FROM pg_stat_bgwriter,
shared_buffers
-----
128MB
(1 row)

buffer_cache_size | percent_dirty
-----+-----
128 MB           |          0.27
(1 row)

training=#

```

- WAL Buffers

Buffer for WAL records before writing to disk.

```

SHOW wal_buffers;
-- Check WAL buffer usage (requires pg_stat_statements extension)
SELECT
    wal_records,
    wal_fpi,
    wal_bytes,
    wal_write,
    wal_sync,
    stats_reset
FROM pg_stat_wal;

```

```

-----
4MB
(1 row)

wal_records | wal_fpi | wal_bytes | wal_write | wal_sync | stats_reset
-----+-----+-----+-----+-----+-----
25659 | 1478 | 9935089 | 774 | 35 | 2025-12-13 19:35:05.209222+03
(1 row)

training=# |

```

- CLOG Buffers (Commit Log)

Stores transaction commit status (in pg_xact directory).

```

-- No direct view command, but can check directory
ls -la $PGDATA/pg_xact/

```

```

root@998-986-02:/home/steve# ls -la $PGDATA/pg_xact/
total 16
drwx----- 2 postgres postgres 4096 Dec 13 19:35 .
drwx----- 19 postgres postgres 4096 Dec 14 11:25 ..
-rw----- 1 postgres postgres 8192 Dec 14 11:50 0000
root@998-986-02:/home/steve#

```

```

-- Estimated size (each transaction uses 2 bits)

```

```
SELECT
    pg_size_pretty((2 * (max_age + 1)) / (8 * 1024 * 1024)) as estimated_clog_size
FROM (SELECT max(age(datfrozenxid)) as max_age FROM pg_database) t;
```

```
estimated_clog_size
-----
0 bytes
(1 row)

training=#
```

- Temp Buffers

Buffers for temporary tables.

```
SHOW temp_buffers;
```

```
temp_buffers
-----
8MB
(1 row)

training=#
```

```
-- Lock table size estimate
SELECT 128 * 1024 as lock_table_size;
```

```
lock_table_size
-----
131072
(1 row)

training=#
```

```
-- Fixed size
-- Shared memory summary
SELECT name, setting, unit FROM pg_settings
WHERE name LIKE '%shared%' AND context = 'postmaster';
```

name	setting	unit
dynamic_shared_memory_type	posix	
min_dynamic_shared_memory	0	MB
shared_buffers	16384	8kB
shared_memory_type	mmap	
shared_preload_libraries		

```
(5 rows)

training=#
```

- Per-Backend Memory
- work_mem

Memory for sort operations and hash tables per operation.

```
SHOW work_mem;
```

```
work_mem
-----
4MB
(1 row)

training=#
```

```
-- Check if queries are spilling to disk
```

```
SELECT query,
       temp_blks_read AS temp_blocks_read,
       temp_blks_written AS temp_blocks_written,
       (temp_blks_read + temp_blks_written) * 8192 AS temp_bytes
FROM pg_stat_statements
WHERE temp_blks_read > 0 OR temp_blks_written > 0
ORDER BY temp_bytes DESC
LIMIT 5;
```

```
query | temp_blocks_read | temp_blocks_written | temp_bytes
-----+-----+-----+-----
(0 rows)

training=#
```

- maintenance_work_mem

Memory for maintenance operations (VACUUM, CREATE INDEX).

```
SHOW maintenance_work_mem;
```

```
maintenance_work_mem
-----
64MB
(1 row)

training=#
```

- Catalog Cache

Cache for system catalog (metadata) queries.

```
-- Catalog cache hits
```

```
SELECT
  sum(relid::bigint) AS relid_sum,
  sum(indexrelid::bigint) AS indexrelid_sum,
  sum(idx_blks_hit) AS cache_hits,
```

```
sum(idx_blks_read) AS disk_reads
FROM pg_statio_user_indexes;
```

```
relid_sum | indexrelid_sum | cache_hits | disk_reads
-----+-----+-----+-----
(1 row)

training=#
```

- Optimizer/Executor Memory

Memory used by query planner and executor.

```
-- Plan cache (prepared statements)
SELECT name, statement, prepare_time
FROM pg_prepared_statements;
```

```
name | statement | prepare_time
-----+-----+-----
(0 rows)

training=#
```

- temp_buffers (Per-backend)

Buffers for temporary tables in each backend.

```
SHOW temp_buffers;
```

```
temp_buffers
-----
8MB
(1 row)

training=#
```

- OS Cache

Operating system page cache (not managed by PostgreSQL).

```
# Linux
free -h
# Cache usage
cat /proc/meminfo | grep -i cache
```

```
# Cache usage
cat /proc/meminfo | grep -i cache
```

	total	used	free	shared	buff/cache	available
Mem:	3.7Gi	774Mi	2.9Gi	19Mi	325Mi	3.0Gi
Swap:	1.0Gi	0B	1.0Gi			
Cached:	272512 kB					
SwapCached:	0 kB					

```
root@998-986-02:/home/steve# |
```

```
# PostgreSQL files in OS cache
vmtouch $PGDATA/base/
```

```
root@998-986-02:/home/steve# vmtouch $PGDATA/base/
Files: 1195
Directories: 5
Resident Pages: 1429/7499 5M/29M 19.1%
Elapsed: 0.079081 seconds
root@998-986-02:/home/steve#
```

PHYSICAL FILES

Data Files

- Table Files

Heap files storing table data.

```
-- Find file location for a table
SELECT pg_relation_filepath('users');
-- Or by OID
SELECT pg_relation_filepath(16384);
-- File size
SELECT pg_size_pretty(pg_relation_size('users'));
```

```
pg_relation_filepath
-----
base/16384/16504
(1 row)

pg_relation_filepath
-----
(1 row)

pg_size_pretty
-----
8192 bytes
(1 row)

training=#
```

- Index Files

B-tree, GiST, GIN, etc. index files.

```
-- Index file location
```

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```
-- Find indexes for 'users' table
SELECT indexname, pg_relation_filepath(indexname::regclass)
FROM pg_indexes
WHERE tablename = 'users';
```

```

indexname | pg_relation_filepath
-----+-----
users_pkey | base/16384/16510
users_email_key | base/16384/16512
(2 rows)

training=#
```

- TOAST Files

Files for large values that don't fit in main table.

```
-- Check if table has TOAST
SELECT relname, reltoastrelid
FROM pg_class
WHERE oid = 'users'::regclass;
-- TOAST file location

SELECT pg_relation_filepath(reltoastrelid) AS toast_file
FROM pg_class
WHERE oid = 'users'::regclass;
```

```

training=# SELECT relname, reltoastrelid
FROM pg_class
WHERE oid = 'users'::regclass;
 relname | reltoastrelid
-----+-----
users    |          16508
(1 row)

training=# SELECT pg_relation_filepath(reltoastrelid) AS toast_file
FROM pg_class
WHERE oid = 'users'::regclass;
 toast_file
-----
base/16384/16508
(1 row)

training=# |
```

- FSM (Free Space Map) Files

Tracks free space in relation files.

```
ls -la $PGDATA/base/*/*_fsm
```

```

root@998-986-02:/home/steve# ls -la $PGDATA/base/*/*_fsm
-rw----- 1 postgres postgres 24576 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/1247_fsm
-rw----- 1 postgres postgres 24576 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/1249_fsm
-rw----- 1 postgres postgres 24576 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/1255_fsm
-rw----- 1 postgres postgres 24576 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/1259_fsm
-rw----- 1 postgres postgres 24576 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/13440_fsm
-rw----- 1 postgres postgres 24576 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/13445_fsm
-rw----- 1 postgres postgres 24576 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/13450_fsm
-rw----- 1 postgres postgres 24576 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/13455_fsm
-rw----- 1 postgres postgres 24576 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/2600_fsm
-rw----- 1 postgres postgres 24576 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/2601_fsm
-rw----- 1 postgres postgres 24576 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/2602_fsm
-rw----- 1 postgres postgres 24576 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/2603_fsm
-rw----- 1 postgres postgres 24576 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/2605_fsm
-rw----- 1 postgres postgres 24576 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/2606_fsm
-rw----- 1 postgres postgres 24576 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/2607_fsm
-rw----- 1 postgres postgres 24576 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/2608_fsm

```

-- FSM size for a table

```
SELECT pg_size_pretty(pg_relation_size('users', 'fsm'));
```

```

training=# SELECT pg_size_pretty(pg_relation_size('users', 'fsm'));
pg_size_pretty
-----
0 bytes
(1 row)

training=# |

```

- VM (Visibility Map) Files

Tracks which pages are known to have only visible tuples.

```
ls -la $PGDATA/base/*/*_vm
```

```

root@998-986-02:/home/steve# ls -la $PGDATA/base/*/*_vm
-rw----- 1 postgres postgres 8192 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/1247_vm
-rw----- 1 postgres postgres 8192 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/1249_vm
-rw----- 1 postgres postgres 8192 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/1255_vm
-rw----- 1 postgres postgres 8192 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/1259_vm
-rw----- 1 postgres postgres 8192 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/13440_vm
-rw----- 1 postgres postgres 8192 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/13445_vm
-rw----- 1 postgres postgres 8192 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/13450_vm
-rw----- 1 postgres postgres 8192 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/13455_vm
-rw----- 1 postgres postgres 8192 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/2600_vm
-rw----- 1 postgres postgres 8192 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/2601_vm
-rw----- 1 postgres postgres 8192 Dec 13 19:35 /var/lib/postgresql/16/main/base/1/2602_vm

```

-- VM size for a table

```
SELECT pg_size_pretty(pg_relation_size('users', 'vm'));
```

```

training=# SELECT pg_size_pretty(pg_relation_size('users', 'vm'));
pg_size_pretty
-----
0 bytes
(1 row)

training=#

```

WAL Files

- WAL Segments

16MB files containing transaction records.

```
ls -lh $PGDATA/pg_wal/
```

```

root@998-986-02:/home/steve# ls -lh $PGDATA/pg_wal/
total 33M
-rw----- 1 postgres postgres 16M Dec 14 12:21 00000001000000000000000002
-rw----- 1 postgres postgres 16M Dec 14 12:10 00000001000000000000000003
drwx----- 2 postgres postgres 4.0K Dec 14 12:10 archive_status
root@998-986-02:/home/steve#

```

-- Current WAL file

```
SELECT pg_walfile_name(pg_current_wal_lsn());
```

```

pg_walfile_name
-----
00000001000000000000000002
(1 row)

```

```
training=# |
```

-- WAL usage

```

SELECT
  pg_size_pretty(pg_wal_lsn_diff(pg_current_wal_lsn(), '0/0')) as total_wal_generated,
  count(*) as wal_files,
  pg_size_pretty(count(*) * 16 * 1024 * 1024) as wal_directory_size
FROM pg_ls_waldir();

```

```

FROM pg_ls_waldir();
total_wal_generated | wal_files | wal_directory_size
-----+-----+-----
33 MB                | 2         | 32 MB
(1 row)

```

```
training=#
```

Configuration Files

- postgresql.conf

Main configuration file.

Chapter 1: PostgreSQL System Architecture

```
-- View all settings
```

```
SELECT name, setting, source FROM pg_settings ORDER BY name;
```

```
| default
exit_on_error          | off
| default
extension_destdir      |
| default
external_pid_file      | /var/run/postgresql/16-main.pid
| configuration file
extra_float_digits     | 1
| default
from_collapse_limit    | 8
| default
fsync                  | on
| default
full_page_writes       | on
| default
geqo                   | on
| default
geqo_effort            | 5
| default
geqo_generations       | 0
| default
```

```
-- Find config file location
```

```
SHOW config_file;
```

```
config_file
-----
/etcd/postgresql/16/main/postgresql.conf
(1 row)

training=#
```

- pg_hba.conf

Host-based authentication file.

```
SHOW hba_file;
```

```
-----
/etcd/postgresql/16/main/pg_hba.conf
(1 row)

training=# |
```

```
-- View current HBA rules
```

```
SELECT * FROM pg_hba_file_rules;
```

rule_number etmask	file_name auth_method options	line_number	type	database	user_name	address
1	/etc/postgresql/16/main/pg_hba.conf peer	118	local	{all}	{postgres}	
2	/etc/postgresql/16/main/pg_hba.conf peer	123	local	{all}	{all}	
3	/etc/postgresql/16/main/pg_hba.conf scram-sha-256	125	host	{all}	{all}	127.0.0.1
4	/etc/postgresql/16/main/pg_hba.conf scram-sha-256	127	host	{all}	{all}	::1
5	/etc/postgresql/16/main/pg_hba.conf peer	130	local	{replication}	{all}	
6	/etc/postgresql/16/main/pg_hba.conf scram-sha-256	131	host	{replication}	{all}	127.0.0.1
7	/etc/postgresql/16/main/pg_hba.conf scram-sha-256	132	host	{replication}	{all}	::1
(7 rows)						

- pg_ident.conf

User name mapping file.

```
SHOW ident_file;
```

ident_file
/etc/postgresql/16/main/pg_ident.conf
(1 row)
training=#

Log Files

- Server Logs

Database server log files.

```
-- Current log file
SELECT pg_current_logfile();
```

(1 row)
training=#

```
-- Log settings
SHOW log_directory;
SHOW log_filename;
SHOW log_rotation_age;
```

```
log_directory
-----
log
1 row)

log_filename
-----
postgresql-%Y-%m-%d_%H%M%S.log
1 row)

log_rotation_age
-----
1d
1 row)
```

- CSV Logs

Logs in CSV format for easy parsing.

```
SHOW log_destination;
```

```
log_destination
-----
stderr,csvlog
(1 row)
```

```
training=# |
```

-- If CSV logging is enabled

```
ls -la $PGDATA/log/*.csv
```

```
root@998-986-02:/home/steve# ls -la $PGDATA/log/*.csv
-rw----- 1 postgres postgres 166 Dec 14 13:52 /var/lib/postgresql/16/main/log/postgresql-2025-12-14_135234.csv
-rw----- 1 postgres postgres 1159 Dec 14 13:52 /var/lib/postgresql/16/main/log/postgresql-2025-12-14_135234.csv.csv
root@998-986-02:/home/steve# |
```

Archive Files

- Archived WAL

Archived WAL segments for backup/recovery.

-- Archive status

```
SELECT archived_count, failed_count FROM pg_stat_archiver;
```

```
archived_count | failed_count
-----+-----
2 | 0
(1 row)
```

```
training=#
```

Control File

- pg_control

Binary file containing critical cluster information.

```
-- Read control file data
SELECT
    checkpoint_lsn,
    redo_lsn,
    redo_wal_file,
    timeline_id,
    full_page_writes,
    next_xid,
    next_oid,
    next_multixact_id,
    next_multi_offset,
    oldest_xid,
    oldest_xid_dbid,
    oldest_multi_xid,
    oldest_multi_dbid,
    checkpoint_time
FROM pg_control_checkpoint();SELECT * FROM pg_control_system();
```

checkpoint_lsn	redo_lsn	redo_wal_file	timeline_id	full_page_writes	next_xid	next_oid	next_multixact_id	n
ext_multi_offset	oldest_xid	oldest_xid_dbid	oldest_multi_xid	oldest_multi_dbid	checkpoint_time			
0/4000028	0/4000028	000000010000000000000004	1	t	0:769	16568		1
0	722	1	1	1	2025-12-14 13:52:34+03			
(1 row)								
[END]								

```
SELECT * FROM pg_control_system();
```

pg_control_version	catalog_version_no	system_identifier	pg_control_last_modified
1300	202307071	7583381970148453164	2025-12-14 13:52:34+03
(1 row)			
training=#			

- Replication Slots

Track WAL retention for replication.

```
SELECT * FROM pg_replication_slots;
```

```

slot_name | plugin | slot_type | datoid | database | temporary | active | active_pid | xmin | catalog_xmin | restart_lsn | confirmed
-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----
(0 rows)

(END)

```

-- Physical slots

```

SELECT slot_name, slot_type, active, restart_lsn
FROM pg_replication_slots
WHERE slot_type = 'physical';

```

```

slot_name | slot_type | active | restart_lsn
-----+-----+-----+-----
test_slot | physical | f      |
(1 row)

```

training=#

- Tablespaces

Alternative storage locations.

```

SELECT spcname, pg_tablespace_location(oid)
FROM pg_tablespace;

```

```

spcname | pg_tablespace_location
-----+-----
pg_default |
pg_global |
ts1 | /var/lib/pgsql/tablespaces/ts1
ts2 | /var/lib/pgsql/tablespaces/ts2
ts3 | /var/lib/pgsql/tablespaces/ts3
ts4 | /var/lib/pgsql/tablespaces/ts4
(6 rows)

```

training=# |

-- Objects in tablespace

```

SELECT relname, relkind
FROM pg_class
WHERE reltablespace = (SELECT oid FROM pg_tablespace WHERE spcname = 'ts1');

```

relname	relkind
pg_toast_16573	t
pg_toast_16573_index	i
customers	r
pg_toast_16584	t
pg_toast_16584_index	i
inventory	r
pg_toast_16595	t
pg_toast_16595_index	i
suppliers	r
shipments	r

(10 rows)

training=# |

▪ Extensions

Optional add-on modules.

```
SELECT * FROM pg_available_extensions;
```

name	default_version	installed_version	comment
pg_trgm	1.6		text similarity measurement and index searching based on trigrams
pg_buffercache	1.4	1.4	examine the shared buffer cache
autoinc	1.0		functions for autoincrementing fields
btree_gin	1.3		support for indexing common datatypes in GIN
citext	1.6		data type for case-insensitive character strings
fuzzystrmatch	1.2		determine similarities and distance between strings
pgcrypto	1.3		cryptographic functions
old_snapshot	1.0		utilities in support of old_snapshot_threshold
dblink	1.2		connect to other PostgreSQL databases from within a database

```
SELECT * FROM pg_extension;
```

oid	extname	extowner	extnamespace	extrelocatable	extversion	extconfig	extcondition
13608	plpgsql	10	11	f	1.0		
16390	pg_buffercache	10	2200	t	1.4		
16398	pg_stat_statements	10	2200	t	1.10		

(3 rows)

training=#

▪ Background Workers

User-defined background processes.

```
-- If pg_background extension is installed
-- All running processes
SELECT pid, backend_type, state, query, query_start
FROM pg_stat_activity;
```

pid	backend_type	state	query	query_start
8377	autovacuum launcher	active	SELECT pid, backend_type, state, query, query_start+ FROM pg_stat_activity;	2025-12-14 14:10:31.654289+03
8379	logical replication launcher			
8411	client backend			
8374	background writer			
8378	archiver			
8373	checkpointer			
8376	walwriter			
(7 rows)				
training=#				

- Query Plan Cache

Cached query plans .

<i>-- Prepared statements act as plan cache</i>		
SELECT name, statement, prepare_time		
FROM pg_prepared_statements;		
name	statement	prepare_time
get_customer	PREPARE get_customer(INT) AS SELECT * FROM customers WHERE id = \$1;	2025-12-14 14:12:15.221362+03
(1 row)		
training=#		